

Marco Mo — Data Scientist
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SUMMARY

Data Scientist with experience in machine learning, NLP, and time-series analysis. Skilled in building end-to-end ML pipelines, handling high-dimensional data, and developing models for real-world applications such as fraud detection and energy forecasting.

PROJECTS

Fraud Detection ML Pipeline (IEEE-CIS)

- Analyzed a highly imbalanced dataset (600k transactions, 3.5% fraud) to identify fraudulent behavior.
- Applied feature engineering and designed validation strategies to prevent data leakage.
- Developed ML models on 400+ features and deployed model-serving APIs with FastAPI.
- Ensured reproducible results via MLflow, and automated workflows with Docker and GitHub Actions.

Health Platform Classification (Bachelor End Project)

- Built NLP-based classification models on 5k+ samples to categorize health transaction platforms.
- Applied active learning to iteratively improve dataset quality and labeling efficiency.
- Evaluated model performance using cross-validation and analyzed results to ensure reliability.
- Identified patterns in text data to support content classification and platform usability.

EXPERIENCE

Teaching Assistant — Data Challenges, Data Analytics, Calculus

Sep 2021 – Jul 2025

Eindhoven University of Technology (TU/e)

- Supported 80+ students weekly, helping them structure analyses and interpret data results.
- Guided student projects and improved clarity in communicating analytical findings.

Technical Lead – Student Team SimEnergy

Sep 2021 – Jul 2023

Eindhoven University of Technology (TU/e)

- Analyzed time-series energy data to model appliance-level consumption patterns.
- Developed machine learning models and evaluated performance across multiple appliance types.
- Communicated insights through dashboards used in energy-focused events.

EDUCATION

MSc Data Science & Artificial Intelligence | Eindhoven University of Technology

2023 – 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration using Different Entropy Measures.

Honors Academy, AI Track | Eindhoven University of Technology

2021 – 2023

- Projects: Energy disaggregation, GAN image generation.

BSc Data Science | Eindhoven University of Technology & Tilburg University

2020 – 2023

- Thesis: Medical AI Platform Text Classification.

SKILLS

Programming: Python, SQL, C++, R, Java

ML & Data: pandas, NumPy, scikit-learn, PyTorch, TensorFlow

LLM / GenAI: LangChain, Ollama, ChromaDB, Qdrant

Data Engineering / MLOps: Spark, MLflow, Docker, CI/CD (GitHub Actions), FastAPI

Visualization: Power BI, Plotly, Dash

Tools: Git

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)